

Identifying Variables — Set A

Level 3 · Ages 11-14 · Name every variable. Give values and units where you can.

Name _____ Date _____

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① Twenty bean seedlings grow in identical pots of the same potting soil on one windowsill, each given 100 mL of water a day. Each week, groups of five get 0, 5, 10 or 15 drops of liquid fertilizer. After four weeks, every plant's height is measured.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ② Students take their pulse for 15 seconds after 2 minutes each of sitting, walking and jogging, resting until it is back to the sitting value in between. Each count $\times 4$ gives beats per minute. (Anyone who should not jog can record.)

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ③ Twenty radish seeds are placed on damp paper towel in each of four identical dishes with lids, kept at 5 °C, 15 °C, 25 °C and 35 °C. Each dish gets 10 mL of water. After five days the seeds that have sprouted are counted.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____